

# Rankine Cycle Practice — Blank Worksheet

Basic 4-state Rankine cycle (no reheat)

|          | State 1        | Pump       | State 2        | Boiler   | State 3        | Turbine    | State 4        |          |
|----------|----------------|------------|----------------|----------|----------------|------------|----------------|----------|
| ISOBARIC | P <sub>1</sub> |            | P <sub>2</sub> | ISOBARIC | P <sub>3</sub> |            | P <sub>4</sub> | ISOBARIC |
|          | T <sub>1</sub> |            | T <sub>2</sub> |          | T <sub>3</sub> |            | T <sub>4</sub> |          |
|          | h <sub>1</sub> |            | h <sub>2</sub> |          | h <sub>3</sub> |            | h <sub>4</sub> |          |
|          | S <sub>1</sub> | ISENTROPIC | S <sub>2</sub> |          | S <sub>3</sub> | ISENTROPIC | S <sub>4</sub> |          |
|          | x <sub>1</sub> |            | x <sub>2</sub> |          | x <sub>3</sub> |            | x <sub>4</sub> |          |

P: kPa • T: °C • h: kJ/kg • s: kJ/kg·K • x: –